// an array to hold the loaded JSON obj arr var animalsJSON = [{ "eng": "rat", "chi": "shu", "pin": "lǎoshǔ", "year": 2020 }, { "eng": "cow", "chi": "niu", "pin": "niú", "year": 2021 }, { "eng": "tiger", "chi": "hu", "pin": "lǎohǔ", "year": 2022 }, { "eng": "rabbit", "chi": "tu", "pin": "tùzi", "year": 2023 }, { "eng": "dragon", "chi": "long", "pin": "lóng", "year": 2024 }, { "eng": "snake", "chi": "she", "pin": "shé", "year": 2025 }, { "eng": "horse", "chi": "ma", "pin": "mǎ", "year": 2026 }, { "eng": "goat", "chi": "yang", "pin": "yáng", "year": 2027 }, { "eng": "monkey", "chi": "hou", "pin": "hóuzi", "year": 2028 }, { "eng": "chicken", "chi": "ji", "pin": "jī", "year": 2029 }, { "eng": "dog", "chi": "gou", "pin": "goǔ", "year": 2030 }, { "eng": "pig", "chi": "zhu", "pin": "zhū", "year": 2031 }];

```
var animals = JSON.parse(animalsJSON);
console.log(animals);
console.log('first animal', animals[0].eng); // rat

// sort (alphabetize) by animal english name
animals.sort((a, b) => (a.eng < b.eng ? -1 : 1))

// animals.sort((a, b) => a.eng - b.eng);
// console.log(animals[0].eng); // chicken (if sort worked)
// animals.sort((a, b) => b.eng - a.eng); // sort A-Z
console.log(animals);
console.log('first animal', animals[0].eng); // chicken

var array = [
    {name: "Pam", age: 44},
    {name: "John", age: 34},
    {name: "Peter", age: 54},
    {name: "Jake", age: 25},
    {name: "Jill", age: 17},
    {name: "Jane", age: 45}
];
console.log(array);
console.log(array[0].age); // 44

var arraySortedByAge = array.sort((a, b) => a.age - b.age); // Sort
youngest first
console.log(arraySortedByAge[0].age); // 19
console.log(arraySortedByAge);

console.log(array);
console.log(array[0].age); // 44

var animalsJSONObj = `{
"rat": { "chi": "shu", "pin": "la&#780;oshu&#780;", "year": 2020 },
"cow": {"chi": "niu", "pin": "niu&#769;", "year": 2021 },
```

```
"tiger": {"chi": "hu", "pin": "la&#780;ohu&#780;", "year": 2022 },
"rabbit": {"chi": "tu", "pin": "tu&#768;zi", "year": 2023 },
"dragon": {"chi": "long", "pin": "lo&#769;ng", "year": 2024 },
"snake": {"chi": "she", "pin": "she&#769;", "year": 2025 },
"horse": {"chi": "ma", "pin": "ma&#780;", "year": 2026 },
"goat": {"chi": "yang", "pin": "ya&#769;ng", "year": 2027 },
"monkey": {"chi": "hou", "pin": "ho&#769;uzi", "year": 2028 },
"chicken": {"chi": "ji", "pin": "ji&#772;", "year": 2029 },
"dog": {"chi": "gou", "pin": "gou&#780;", "year": 2030 },
"pig": {"chi": "zhu", "pin": "zhu&#772;", "year": 2031 }
}`;

var animalsObj = JSON.parse(animalsJSONObj);
// alert(animalsObj.year);
console.log(animalsObj);
console.log(animalsObj.chicken);
// chicken: {chi: 'ji', pin: 'ji&#772;', year: 2029 }
console.log(animalsObj.cow.chi);  // niu
console.log(animalsObj.tiger.pin); // la&#780;ohu&#780;
console.log(animalsObj.rabbit.year); // 2023
console.log(animalsObj.chi); // undefined
console.log(animalsObj.pin); // undefined
console.log(animalsObj.year); // undefined

    // AJAX LOAD JSON
    // function loadJSON() {

    //     // 1.) instantiate an AJAX Obj
    //     const xhr = new XMLHttpRequest()

    //     // 4.) respond to server side events
    //     xhr.onreadystatechange = function() {

    //         // 5.) if data is loaded..
    //         if(xhr.readyState == 4 && xhr.status == 200) {

    //             // 6.) do something with loaded data
    //             animals = JSON.parse(xhr.responseText)
    //             outputAnimals()

    //         } // end if

    //     } // xhr.onreadystatechange

    //     // 2.) open a server connection
    //     xhr.open("GET", "json/zodiac-animals.json", true)

        // 3.) send for the data
        //xhr.send()

    //} // end function loadJSON()
```

```
    // runs once zodiac-animals.json is loaded, parsed and passed to
animals array
    function renderAnimals(arr) {

        const output = document.getElementById("output")
        // make a table one row per animal
        // but first, test to see if we have our data..
        // output.innerHTML = '<h1>' + animals[1].chi + '</h1>'

        var tbl = `<table width="100%" border="0px"
                    cellpadding="5px" align="center"
                    style="background-color:#FFF">`

        tbl += `<tr style="background-color:#CCC;
                    padding:10px; font-size:1.75rem">
                    <td colspan="5" align="center">
                        <h2 style="color:#777">Chinese Zodiac Animals</h2>
                    </td>
                </tr>`

        tbl += `<tr style="background-color:#777; color:#DDD;
                    padding:10px; font-size:1.5rem">
                    <th>Animal</th>
                    <th>Pinyin</th>
                    <th>Character</th>
                    <th>Sound</th>
                    <th>Years</th>
                </tr>`

        // a loop to make 12 table rows, one per animal
        for(let i = 0; i < arr.length; i++) {

            tbl += `<tr>

                        <td>
                            <img src="images/animals/${arr[i].eng}.jpg"
                                width="125px" height="auto">
                            <br/>
                            <h3>${arr[i].eng}</h3>
                        </td>

                        <td>
                            <h2>
                                ${arr[i].pin}
                            </h2>
                        </td>

                        <td>
                            <img src="images/chars/char-${arr[i].chi}.jpg"
                                width="100px" height="auto">
```

```
                        </td>

                        <td>
                            <img src="images/sound-icon.png" width="35px"
height="auto"
                                    onclick="playSound()"
id="${arr[i].eng}" style="cursor:pointer">
                        </td>
                        <td>
                            <h4 style='text-align:left'>
                                ${calculateYears(arr[i].year)}
                            </h4>
                        </td>
                    </tr>`

        } // end for loop

        tbl += `</table>`

        output.innerHTML = tbl

    } // function renderAnimals()

    // plays when any of the 12 sound icons is clicked
    var sound = new Audio()

    function playSound() {

        // play the correct sound for the clicked animal
        sound.src = "audio/" + event.target.id + ".mp3"
        sound.play()

    } // function playSound()

    function calculateYears(year) {

        let years = "" // for concatenating years: "1988, 2000, 2012"
        // a loop that takes the active year, backs up a few centuries,
and then runs a century or so into the future
        let startYr = year - 264 // back up 20 zodiac cycles
        let endYr = year + 156

        for(let i = startYr; i <= endYr; i+=12) {
            if(i == year) { // if it's the active year, make it red
                years += '<span style="color:red">' + i + '</span>, '
            } else if(i == endYr) { // if last year don't add a comma
                years += i
            } else { // if it's neither of the above, add a comma
                years += i + ", "
            }
```

```
        }
        return years
    }

    renderAnimals(animals);
```